VERDANT

Tulips IOT – Environmental Requirements

Table Summary

Parameter	**Optimal Range**	**Notes**
Temperature	4°C – 16°C (day), 5°C – 10°C (night)	Avoid above 20°C; causes tulip topple disorder
Relative Humidity	60% – 80%	Above 80% risks *Botrytis tulipae* and flower blast
Soil type	Well-draining loamy or sandy loam	Avoid clay; amend with coarse sand and compost
Soil Moisture Content	50% – 70% of field capacity	Moist but never waterlogged; check top 5–10 cm
Soil pH	6.0 – 7.0 (ideal: 6.5)	Slightly acidic to neutral
Sunlight Exposure	6 – 8 hours per day	Full direct sun; keep greenhouse covering clean

Tulip Environmental Requirements

1. Optimal Temperature Range

The ideal greenhouse temperature for tulip cultivation is generally between 4°C and 16°C, and temperatures outside the optimal range increase disease risk. Temperatures above 20°C should be avoided as they can trigger tulip toppling disorder (Meir, 2023).

2. Optimal Relative Humidity Range

The relative humidity should be kept between 60% and 80%, measured just above the crop, and excessive humidity increases the risk of *Botrytis tulipae* and flower damage (Forcing tulips: A practical guide., 2020).

3. **Recommended Soil Type**

The Agriculture Institute states that sandy loam or loamy soils with a pH between 6.0 and 7.0 are ideal for tulips, as heavy clay soils should be avoided due to poor drainage, though they can be improved with compost or coarse sand (*Growing Tulips*, 2025).

4. **Optimal Soil Moisture Content**

Cornell's Flower Bulb Research Program conducted experiments using soil moisture levels of 15%, 30%, 45%, and 60% of container capacity, providing empirical data on how moisture levels affect tulip root development. Their recommendation is a soil moisture content of 50 – 70% of field capacity (*Soil Moisture and Tulip Root Growth in the Cooler | Flower Bulb Research Newsletter*, 2018).

5. **Optimal Soil pH Range**

Tulip bulbs require well-drained soil with a pH of 6.0 to 6.5, and raised beds or sandy soil amended with organic matter provide an advantage for drainage. The Plant Aide further explains that this slightly acidic to neutral pH range of 6.0 to 7.0 ensures optimal nutrient availability, particularly for micronutrients like iron and manganese (*Planting & Growing Tulips*, 2018).

6. **Suitable Hours of Sunlight Exposure**

Tulips require full sun for the best display, meaning at least 6 hours of bright, direct sunlight per day, and they prefer fast-draining soil. Without adequate light, tulips suffer from etiolation, a condition where stems become abnormally elongated and spindly as the plant stretches toward light sources (“Do Tulips Need Full Sun?,” 2026).

References

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